Composable retrieval and reasoning mechanisms

Mechanisms can be combined. Panel order does not imply replacement or a performance ranking.

Retrieve then generate

Retrieve passages, then condition answer generation.

RAG (Lewis et al., 2020)
Triple-Fact Retriever; DR-RAG

Structure-aware retrieval

Preserve sections and document relationships.

PdfTriage; HiQA
DocReLM

Graph-based retrieval

Connect entities or evidence within and across documents.

GraphRAG; LightRAG
HyperGraphRAG

Visual retrieval

Index page or region image representations.

ColPali; VisRAG
M3DocRAG

Agent control

Choose retrieval / tool actions; revise the evidence context.

MA-RAG; CogDoc
AutoThinkRAG

Combine, then evaluate

Example: visual index + graph + iterative retrieval control.

Report answer / evidence quality and token, latency, tool costs.